

AnalystLab Africa Internship Program

E-Commerce Data Analysis Report

Week 1-2 – Data Cleaning and Exploratory Data Analysis

Dataset – Online Retail Transaction Dataset

Tools Used – Python (Pandas, Matplotlib, Seaborn) and MySQL

Dataset Overview

The dataset contains transactions from a UK-based E-commerce business. After cleaning, the dataset contained 392,692 transactions, 4,338 unique customers, 3,877 unique products and covered 37 countries between December 2010 and December 2011.

Key Insights

1. UK Market Dominance - The UK accounts for 82% of total revenue (£7,285,025 out of £8,887,208). This tells us that the business is heavily dependent on one market.
2. Seasonal Sales Pattern - Revenue peaked in November 2011 at £1,156,205, which is almost double the average monthly revenue. This suggests strong Christmas season demand.
3. Top Product Category - Best selling products are home décor, craft and gift items suggesting that the business serves a specific niche market.
4. VIP Customer Concentration - The top 10 customers contributed disproportionately large revenue. Customer 14646 alone spent £280,206.
5. Low Average Transaction Value - The average order value is £22.63 suggesting the business relies on high volume of small purchases rather than big ticket items.

Conclusion

This analysis shows a healthy e-commerce business with a strong UK presence, clear seasonal demand patterns and a loyal customer base. Future analysis should focus on customer retention strategies as well as international market expansion.